

Transformer-Based Gas Distribution Mapping with Sparse and Biased Sensor Observations

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Abstract—Gas distribution mapping (GDM) is essential for industrial safety, environmental monitoring, and emergency response. Traditional methods rely on dense and uniform sensor deployments, which are costly, inflexible, and often impractical in dynamic or unknown environments. Mobile robotic platforms offer a promising alternative to perform flexible and efficient gas sensing. Yet, robot-assisted GDM has to deal with the sparse, irregular sampling positions, which hinder the performance of conventional interpolation techniques and many learning-based models that assume fixed input structures. In this work, we propose a transformer-based model for GDM that reconstructs gas distribution and wind fields from limited, uneven sensor inputs by leveraging self-attention and Fourier-based positional encoding to capture long-range spatial dependencies and adapts to biased sampling position layouts. Trained on simulated CFD data and evaluated on both simulated ground-truth and real-world wind tunnel measurements, our approach consistently outperforms linear interpolation, Kernel DM+V, and CNN baselines. Under highly sparse conditions (e.g., 20 samples), it achieves up to 98.3% structural similarity index and 0.0005 mean squared error, significantly outperforming other methods. It also exhibits strong robustness in low-observation regimes and is capable of reconstructing gas plumes with low-cost, slow-recovery metal-dioxide sensor measurements in a complex environment. This work highlights the transformer’s potential for robot-assisted GDM tasks in challenging sensing conditions.

Keywords

Robotic Olfaction, Gas Distribution Mapping, Transformer Model, Environmental Monitoring

Highlights

Transformer reconstructs gas and wind fields from sparse, irregular observations.

Fourier positional encoding improves robustness to sparse and biased sensor layouts.

Transformer surpassed current baselines in a comparative evaluation on simulated gas distribution data.

Wind-tunnel validation shows robustness to noisy MOX sensor measurements.

I. Introduction

Monitoring and analyzing gas emissions is essential across a range of domains, including industrial manufacturing [1], environmental monitoring [2], and risk assignment [3]. In these applications, capturing not only the presence of gas but also its spatial distribution is critical for detecting leaks, assessing exposure risks, and informing mitigation strategies. Gas Distribution Mapping (GDM), which aims to reconstruct spatial concentration fields from sensor data, can play an important role in improving situational awareness and supporting decision-making. Conventional Gas Distribution Mapping (GDM) approaches deploy dense networks of stationary gas sensors [4]. While these sensor networks can provide detailed data, they are expensive to install and maintain, and they are impractical to deploy in unknown or highly dynamic environments. These limitations motivate the exploration of new GDM methods that can achieve high spatial resolution with minimal sensor deployment.

Mobile robots have emerged as a promising platform to carry out gas distribution mapping due to their mobility and flexible sampling capabilities [5]–[7]. A single robot-mounted gas sensor can collect measurements over a wide area by scanning through the environment or adaptively inspecting spots, reducing the need for extensive fixed sensor installations. However, existing robot-assisted GDM techniques typically assume that the robot can densely and uniformly sample the environment. In practice, various factors (such as obstacles, confined spaces, or safety restrictions) often prevent robots from accessing certain locations, resulting in inherently sparse and unevenly distributed measurements. This sparse and irregular sampling poses significant challenges to mapping accuracy, as unsampled regions lead to large uncertainties in the reconstructed gas distribution, which calls for advanced reconstruction methods that are capable of producing reliable gas distribution maps with limited and unevenly spaced observations.

Beyond the spatial distribution of samples, the physical

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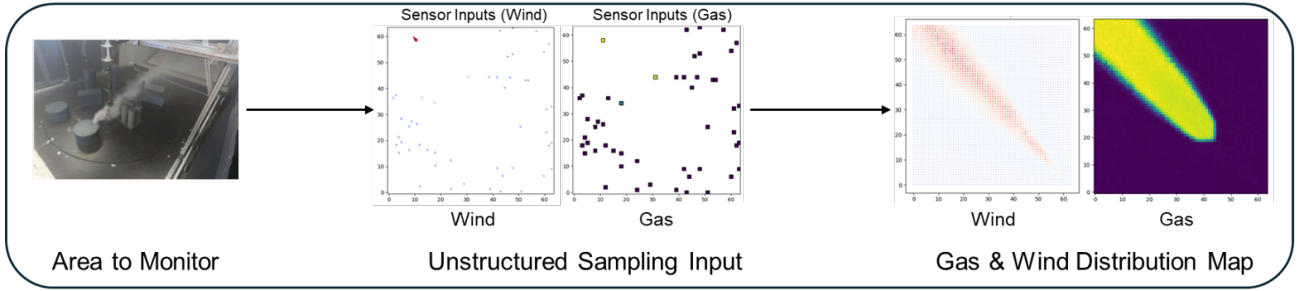


Fig. 1: Schematic diagram of an real-world gas distribution mapping problem setup: the sampling positions are not optimized for reconstructing the gas distribution, resulting spatially unstructured measurement inputs for a learning model.

process underlying gas dispersion also plays a critical role in gas distribution mapping. In realistic scenarios, accounting for airflow can substantially improve gas distribution mapping, as gas molecules are transported through wind-driven turbulent advection, which creates chaotic plumes and intermittent patches of high concentration [8]. Traditional approaches like Computational Fluid Dynamics (CFD) attempt to solve the governing fluid equations for advection–diffusion, but they are computationally expensive. Additionally, as a typical model-driven approach, their strong reliance on precise knowledge of boundary conditions of the target areas limits their adaptability to unfamiliar environments [8]. Instead, several works integrate wind information directly into the mapping process. For example, a statistical gas mapping method (Kernel DM+V/W) shapes each sensor’s influence according to the local wind vector, effectively encoding where a sensed gas patch likely came from and where it will travel [8], [9]. Similarly, joint estimation frameworks have been proposed to simultaneously map gas concentration and wind flow, exploiting the strong correlation between these two fields to infer gas concentrations in unsampled areas [10], and indicated that even including sparse wind measurements can improve the quantitative agreement between the estimated and actual gas concentration fields, particularly in complex environments with obstacles or turbulent plume behaviour [10]. Beyond physics-based and data-driven models, learning-based approaches have emerged as a promising solution for incorporating airflow into GDM. Machine learning methods can learn the underlying relationship between limited sensor observations and the overall fluid field, offering adaptability to new environments without relying on predefined conditions. For instance, physics-informed neural networks (PINNs) can fuse sparse pointwise velocity measurements with partial differential equations, enabling reconstruction of a flow field with higher accuracy than a traditional RANS solver under the same sparse-data conditions [11]. Generative models further demonstrate the potential of learned fluid representations: a denoising diffusion model was shown to capture complex flow patterns from simulation data and

produce accurate fluid field predictions without explicit physics knowledge [12], while GAN-based models can super-resolve coarse fluid simulations by a factor of 8 in each spatial dimension to recover fine-grained flow details [13], which allows to learn the underlying relationships between sampling points and overall fluid fields that support point-to-point reconstruction with sparse and unbalanced observations. These AI-driven techniques do not outright replace classical CFD; rather, they offer faster, more flexible tools to estimate wind and gas distributions when real-time computation or incomplete information is a limiting factor. A learning model can leverage sparse and unevenly distributed gas and airflow measurements to effectively compensate limited spatial sampling coverage is promising to real-world applications.

In addition to modelling physical airflow dynamics, gas distribution mapping in robotic application must contend with the structural challenges posed by real-world gas measurements. First, the number and locations of sampling points are unfixed, making rigid network architectures such as multilayer perceptron unsuitable. Second, the distribution of gas concentration within the region is highly sparse, with most of the environment containing near-zero readings outside localized plumes. demands that the model effectively extract informative patterns from predominantly low-value inputs. Compounding this, robotic motion often results in uneven spatial sampling, where some regions are densely observed while others remain underexplored. To produce accurate gas maps under these conditions requires a model capable of encoding the relative positional information of the inputs to capture spatial correlations between samples, and generalizing across highly imbalanced spatial coverage.

In this work, we propose a transformer-based model for gas distribution mapping in robotic applications. The attention mechanism allows the model to capture long-range spatial correlations [14], which is essential for inferring structure from sparsely sampled environments. To encode spatial relationships, we apply positional encoding to the sampling coordinates, allowing the model to interpret the relative locations of sensor inputs. This combination makes

the transformer architecture well-suited for reconstructing fluid fields with irregular and unstructured inputs. Given that near-ground airflow is primarily horizontal due to gravity and surface friction, we formulate the mapping problem in two dimensions, aligned with the robot’s operational plane. Each input sample comprises a tuple of gas concentration, wind velocity components (in the x and y directions), and its spatial coordinates. After applying positional encoding to the coordinates and feature encoding to the sensor readings, the inputs are fused and processed through an encoder–decoder architecture, which outputs two continuous fields: gas concentration and wind distribution, via separate output heads.

To train the model, we generated a dataset using computational fluid dynamics (CFD) simulations of gas dispersion under diverse conditions. These simulations include varying source locations, emission intensities, and wind directions to ensure coverage of plausible real-world scenarios. After training, the model was validated using real-world data collected from a low-speed wind tunnel with a mobile gas sensing platform. Results show that our approach can accurately reconstruct the gas concentration and wind field across the area from sparse and unevenly distributed sensor samples. Notably, the approach remains effective even in the presence of high sensor noise and baseline drift, consistently identifying the regions influenced by gas plumes.

The main contribution of this work is the development of a transformer-based model for gas distribution mapping that can accurately reconstruct gas concentration and wind fields from sparse and irregular sensor observations. By combining attention mechanisms and positional encoding, the model effectively captures long-range spatial correlations and encodes sampling geometry, making it particularly well-suited for robotic GDM tasks. Extensive evaluations with both simulation and real-world dataset demonstrate the robustness and practical utility of the proposed approach, offering an efficient and practical solution to gas distribution mapping in complex, partially observed sensing conditions.

II. Related Work

Gas distribution mapping (GDM) methods are generally divided into model-based analytical, statistical and learning-based approaches. Traditional methods estimate concentration fields using explicit physical models, probability estimates [15], [16] or interpolation of gas sensor data [17], which are straightforward to use but often assume dense, uniform sensor coverage. This inflexibility has motivated the development of learning-based approaches that train machine learning models to predict gas distributions. They can capture more complex plume patterns, but they generally require large training datasets and may struggle to generalize when sample sizes are limited or sensor readings are irregular and noisy. As a result, the current approaches cannot fully address the challenges of

varying sample densities, sparse and uneven sampling, or deployment in complex real-world environments.

A. Statistical and Interpolation Methods

Early GDM approaches use spatial inference to extrapolate sparse measurements. For instance, kernel-based estimators (e.g. Kernel DM+V/W [16]) predict concentration at a point from weighted nearby readings. Likewise, Gaussian process regression has been applied to model plume fields: Stachniss et al. propose a sparse GP mixture that captures both the smooth background and localized high-concentration patches of a plume [18]. Other works use probabilistic filters, such as Blanco et al. introduced a Bayesian Kalman filter on a grid, estimating both mean concentration and variance [15]. These methods typically assume a (quasi)-static distribution and dense sensor coverage. They can produce coarse maps of the average concentration but require many readings and struggle with highly non-uniform sampling. For example, Blanco et al. demonstrate how a Kalman-based model can integrate wind data and quantify uncertainty, but note that it assumes a largely time-invariant plume [15]. In brief, statistical GDM methods can efficiently interpolate and quantify uncertainty under ideal sampling, but their accuracy degrades if sensors are sparse, uneven, or the plume is rapidly changing.

B. Physics- and Model-Based Methods

Alternatively, physics-based models estimate gas dispersion by explicitly physics-based models estimate gas dispersion by solving partial differential equations (PDEs) to simulate airflow and contaminant transport under known environmental conditions. Classic Gaussian plume models assume a steady point source and constant wind to compute concentration analytically. In complex or confined geometries, Computational Fluid Dynamics (CFD) is used to directly model gas dispersion. By numerically solving the Navier–Stokes and advection-diffusion equations, CFD can predict gas dispersion with fine detail. For example, Osenbroch and Hjertager note that while empirical plume models are fast and simple, realistic hazard assessments often require CFD for accuracy in cluttered environments [19]. CFD captures obstacles and turbulence, but is computationally intensive and inflexible for on-the-fly mapping. Hybrid simulation models have also been used: Monroy et al. build a gas dispersal simulator combining CFD (e.g., OpenFOAM) with a filament-based plume model to generate realistic gas plumes [20]. In practice, physics-based methods yield high-fidelity dispersion predictions under known conditions (source parameters, boundary conditions), but they require detailed environmental information and cannot easily adapt to unknown or changing conditions in real time or complex industrial settings.

C. Learning-based Gas Distribution Mapping

Learning-based approaches using machine learning models to reconstruct concentration fields or support tasks such as probabilistic source localization and informative planning, have been motivated by the limitations of analytical plume models that are often oversimplified or computationally prohibitive, or statistical/interpolation methods, which assume local smoothness and work best with dense, uniform sampling. Learning-based models are promising in capturing complex nonlinear plume patterns and fusing multi-modal inputs such as wind and environment geometry, though they require representative data and careful design to generalize beyond training distributions. For example, Convolutional neural networks (CNNs) have been applied as powerful interpolators. Winkler et al. introduce the Gas Distribution Decoder, a deep CNN that upsamples sparse sensor inputs to a high-resolution gas map [21]. On synthetic and real data, this method significantly outperforms classical interpolation, especially in capturing plume edges and shapes. Other models incorporate spatiotemporal learning: Winkler et al. propose a hybrid convolutional-LSTM network that processes sequences of past readings. They show that exploiting temporal correlations of plume dynamics yields more accurate maps than static models [22]. Beyond purely data-driven field interpolation, some works integrate learned plume surrogates directly into probabilistic source localization frameworks. Jin et al. propose a Data-Driven Plume Model (DDPM), trained on CFD simulations that conditions on obstacle maps and hypothesized source locations to predict steady-state concentration fields [23]. The DDPM is coupled with a Bayesian Source Term Estimation algorithm, enabling robust map-aware inference for gas source localization in cluttered built environments where pseudo-Gaussian models fail. Experiments in simulation and wind-tunnel setups showed improved accuracy and convergence compared to analytic baselines, though performance was sensitive to strongly unsteady plume dynamics and sensor limitations. In general, deep learning methods can model complex nonlinear dispersion patterns without handcrafted physics. However, they typically require large, representative datasets and are less reliable when training samples are limited or biased. For instance, CNNs have a limited receptive field and expect fixed-size (often grid) inputs, which makes them less suited to highly irregular sensor layouts. The limited receptive field of CNNs also constrains their ability to capture global features.

More recently, physics-informed neural networks [11] (PINNs) have emerged, which enhance prediction reliability and robustness by incorporating partial differential equations into the training process. However, the solution space of fluid equations is highly nonlinear, requiring more comprehensive and high-quality data to achieve accurate modelling [24]. Unfortunately, there are no specific

large-scale open-source datasets available for GDM tasks, and self-collected datasets are often limited in size or contain noise. As a result, PINNs are prone to falling into trivial solutions due to the inconsistency between the gradients of the physics-based loss and the data-driven loss. By contrast, newer attention-based models can better handle variable inputs. Due to its adaptability to variable-length inputs and its ability to generate high-dimensional representations from low-dimensional inputs through the attention mechanism, the Transformer is considered a promising model for fluid-related problems. Notably, Santos et al. develop Senseiver, an attention-driven framework that encodes arbitrary sets of sparse observations into a latent representation and reconstructs the full field [25]. The Senseiver uses cross-attention to fuse sensor values at arbitrary locations and has shown state-of-the-art reconstruction accuracy in several fluid-flow benchmarks. These results suggest that Transformer-like architectures, which flexibly aggregate sparse data and learn global correlations, are a promising direction for GDM. Transformer-based approaches have already been applied to various fluid dynamics problems, such as turbulence reconstruction [26] and temporal flow prediction [27].

In summary, deep learning approaches offer strong adaptability and can capture complex plume structures, but their effectiveness depends heavily on the quality and quantity of training data, as well as thoughtful model design. Traditional methods like statistical estimators and interpolation are simple and interpretable, yet they struggle under sparse or uneven sampling conditions. Physics-based simulations, while capable of producing accurate results under well-characterized conditions, are computationally intensive and often impractical for real-time applications. Addressing real-world gas distribution mapping requires to balance the trade-offs between model complexity, computational cost, interpretability, and robustness to data limitations. In the scope of this work, we are interested in Transformers as they are particularly well-suited to sparse and irregular inputs due to their ability to model long-range dependencies and process variable-sized input sets using attention mechanisms.

III. Materials and Methods

This section outlines the full pipeline used to develop and evaluate a transformer-based model for gas distribution mapping under sparse and irregular sensor observations. We first describe the generation of a synthetic dataset using computational fluid dynamics (CFD) in Section III-A. This dataset provides diverse ground truth gas and wind fields under varying source and airflow conditions. From these simulations, we apply probabilistic sampling strategies to simulate realistic, unbalanced sensor inputs. We then introduce the proposed transformer-based model architecture in Section III-B. This model is designed to process variable-sized inputs through positional encoding and self-attention, enabling it

to capture long-range spatial dependencies. Corresponding training procedure is detailed, including regularization strategies and a curriculum learning approach to improve model robustness across different input densities. Next, in Section III-C, we introduce the validation of the trained model using a real-world wind tunnel dataset to assess its extrapolation capability. For performance comparison, we benchmark our work against classical and learning-based baseline approaches using quantitative metrics under realistic conditions. Corresponding definitions of the evaluation criteria and the setups of the test data to reflect different levels of input sparsity and sampling imbalance can be found in Section III-C.

A. Training Data Generation and Downsampling

Gas dispersion data were generated using the open-source CFD toolbox OpenFOAM [28]. The simulation domain (Fig. 2) was a 2D square region of 1 m $\times$ 1 m, discretized with a structured mesh of 100 $\times$ 100 uniform cells. Each case included a randomly located gas source and varying airflow velocities from two orthogonal inlets.

1) Flow Simulation Setup: The airflow was simulated as a two-dimensional, steady, incompressible turbulent flow. The governing equations are the Reynolds-Averaged Navier-Stokes (RANS) equations with the k - ε turbulence model:

$$\nabla \cdot \vec{u} = 0 \quad (1)$$

$$\rho \vec{u} \cdot \nabla \vec{u} = -\nabla p + \mu \nabla^2 \vec{u} + \nabla \cdot (\rho \overline{u'u'}) \quad (2)$$

Here, $\vec{u} = (u, v)$ is the mean velocity vector, ρ is the fluid density, p is the pressure, μ is the dynamic viscosity, and $\overline{u'u'}$ represents the Reynolds stress tensor due to turbulence.

The k - ε model introduces two transport equations for turbulent kinetic energy k and its dissipation rate ε :

$$\frac{\partial k}{\partial t} + \vec{u} \cdot \nabla k = P_k - \varepsilon + \nabla \cdot \left[\left(\nu + \frac{\nu_t}{\sigma_k} \right) \nabla k \right] \quad (3)$$

$$\frac{\partial \varepsilon}{\partial t} + \vec{u} \cdot \nabla \varepsilon = C_{1\varepsilon} \frac{\varepsilon}{k} P_k - C_{2\varepsilon} \frac{\varepsilon^2}{k} + \nabla \cdot \left[\left(\nu + \frac{\nu_t}{\sigma_\varepsilon} \right) \nabla \varepsilon \right] \quad (4)$$

In these equations:

- k Turbulent kinetic energy, representing the energy contained in eddies per unit mass.
- ε Dissipation rate of turbulent kinetic energy.
- P_k Production term of k due to velocity gradients.
- ν Molecular (kinematic) viscosity of the fluid.
- ν_t Eddy viscosity representing turbulent diffusion.
- $\sigma_k, \sigma_\varepsilon$ Turbulent Prandtl numbers for k and ε , controlling their diffusion rates.
- $C_{1\varepsilon}$ Empirical constant governing the generation of ε from k (typically around 1.44).
- $C_{2\varepsilon}$ Empirical constant representing the destruction of ε (typically around 1.92).

2) Boundary Conditions and Source Setup: Table I summarizes the boundary conditions. Two air inlets are placed along the left and bottom boundaries. Each inlet velocity is randomly sampled from a uniform distribution between 0.1 and 0.5 m/s. The gas source is randomly placed within the computational domain of size 1.0 m $\times$ 1.0 m for each simulation run. Its center coordinates (x, y) are independently and uniformly sampled from $[0, 1.0]$ m in both directions. The emission strength Q is also uniformly drawn from a predefined range of 0.05 kg/s to 0.2 kg/s. This uniform randomization of both location and intensity ensures diverse leak scenarios and plume patterns, facilitating robust model training and evaluation.

TABLE I: Boundary Conditions for Flow Simulation

Boundary	Type	Details
Left	Velocity Inlet	$u \sim U(0.1, 0.5)$ m/s
Bottom	Velocity Inlet	$v \sim U(0.1, 0.5)$ m/s
Right	Pressure Outlet	Zero gradient
Top	Pressure Outlet	Zero gradient
Interior	Gas Source	Random position & intensity

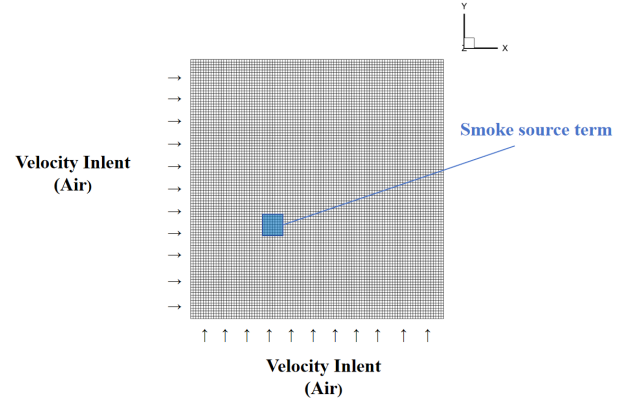


Fig. 2: Simulation domain setup and mesh layout

3) Synthetic Sampling from the Simulated Ground Truth: Each simulation case corresponds to a unique gas dispersion scenario, defined by a specific combination of gas source placement, emission rate, and airflow conditions. For each case, the ground truth gas concentration and wind fields are interpolated onto a uniform 64 $\times$ 64 grid. From these maps, N points are randomly sampled to simulate sensor inputs, with $(N = 20, 50, 100 \dots)$. Each sample includes the spatial coordinates (x, y) , gas concentration c , and wind velocity components (u, v) .

The dataset comprises 400 cases, divided into training, validation, and testing sets in a 3:2:2 ratio. To account for variability in sensor deployment, each case undergoes 16 independent sampling trials, resulting in different sparse input configurations. To simulate unbalanced sensor distributions, we adopt a gas-concentration-weighted sampling

strategy. Specifically, the probability p_i of sampling a given cell i is computed according to:

$$p_i = \frac{(c_i + \varepsilon)^\alpha}{\sum_{j=1}^N (c_j + \varepsilon)^\alpha} \quad (5)$$

where c_i is the gas concentration at cell i , $\varepsilon = 0.01$ is a small constant for numerical stability, and α controls the sampling bias: $\alpha > 0$ favours high-concentration regions, $\alpha < 0$ favours low-concentration regions, and $\alpha = 0$ results in uniform sampling. This formulation allows us to simulate a range of realistic sensor deployment patterns, from plume-focused sampling to uniform or background-biased configurations.

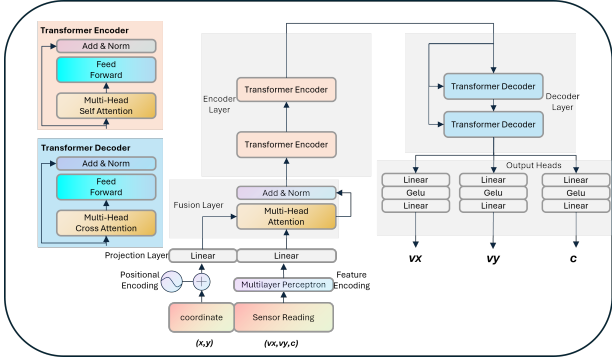


Fig. 3

B. Transformer-Based GDM Model and Training

We design a Transformer-based model, as depicted in Fig. 3, to reconstruct gas concentration and wind fields from sparse and irregular sensor observations. Each input sample includes spatial coordinates (x, y) , wind speed components (u, v) , and gas concentration (c) . To encode these inputs, the model incorporates two parallel embedding modules: a Dynamic Positional Encoder that processes spatial locations, and a Feature Encoder that processes gas and wind measurements.

The dynamic positional encoder applies Fourier-based encoding to the (x, y) coordinates, followed by a multi-layer perceptron (MLP) to generate a 128-dimensional embedding. This representation captures high-frequency spatial patterns (e.g., fluctuating gas concentration levels) and promotes generalization across varying sensor layouts. In parallel, the feature encoder processes the (u, v, c) measurements through a two-layer MLP with GELU (Gaussian Error Linear Unit) [29] activation and dropout, producing a feature embedding of the same dimension. Both the position and feature embeddings are projected into a shared latent space and fused using multi-head self-attention. A residual connection and LayerNorm are applied to stabilize training and preserve spatial structure. The fused embeddings are passed through a two-layer Transformer encoder to produce a global context representation.

To reconstruct the full spatial fields, we generate a set of decoder queries representing each cell of the target 64×64 output grid. These queries are constructed by combining fixed positional encodings of the grid coordinates with a learned latent query. The decoder queries attend to the encoder memory via a two-layer Transformer decoder, enabling the model to generate a global feature representation. Finally, three parallel MLP heads regress the global feature into spatial maps of gas concentration c and wind velocity (u, v) , respectively. The resulting maps of c , u and v are reshaped into a 64×64 output.

The model is trained using the mean squared error (MSE) loss function, with L2 regularization [30] applied to prevent overfitting. We adopt the OneCycleLR strategy [31] for dynamic learning rate scheduling to facilitate faster convergence and stable optimization. To further enhance training efficiency and model robustness, we employ a curriculum learning strategy [32]: the model is initially trained on samples with 100 sensor inputs and then on sparser cases with 50 inputs to improve generalization under reduced sensing conditions.

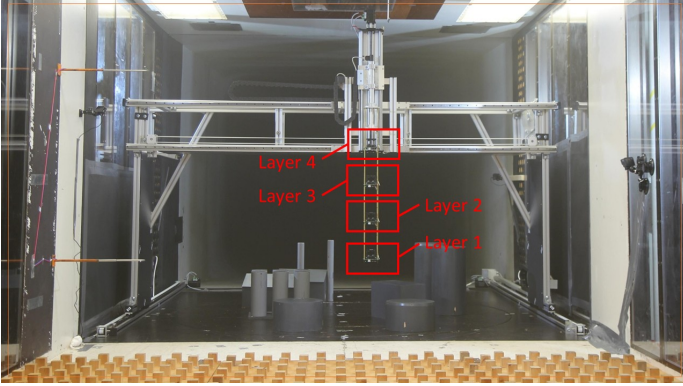
C. Validation on Real-world Wind Tunnel Data

The validation of our model is based on an open-source, real-world dataset collected in a low-speed wind tunnel located in Marknesse, Netherlands [33]. The experimental area measures $2.40 \times 2.40 \times 0.88 \text{ m}^3$ and contains a scaled-down industrial site model with obstacles and a gas source emitting propylene glycol vapour (Figure 4a). As shown in the Figure 4b, a robotic arm carries four gas sensing units positioned at different vertical heights (i.e., at Layer 1, Layer 2, Layer 3 and Layer 4), acquiring measurements across a 20×20 horizontal grid in the XY-plane. Each sensing unit comprises a MiCS5524 metal oxide (MOX) gas sensor, a PID-AH2 photo-ionization detector (PID) for measuring gas concentrations and a Trisonica Mini ultrasonic 3D anemometer for horizontal wind velocity. As one can observed in Figure 4c, compared to the PID, which provides calibrated and relatively sharp gas concentration peaks and fast recovery, the MOX sensor responses are characterized by slower recovery dynamics, susceptibility to baseline drift, and did not report calibrated concentration levels. This experimental setup yields a high-fidelity dataset that reflects realistic dispersion conditions with complex airflow structures and spatial obstructions.

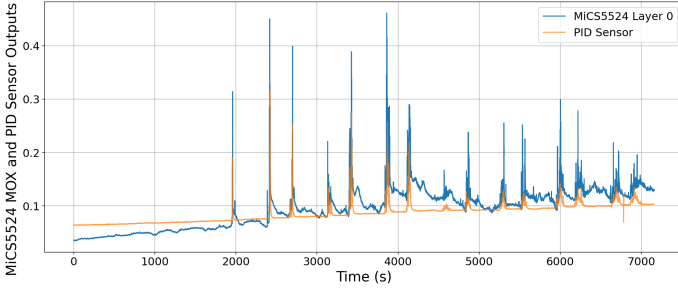
For the evaluation of the learning models, we select the measurements collected from Layer 1, the plane closest to the gas source, where gas signal intensity and spatial variation are most prominent. Specifically, the MiCS5524 MOX readings from this layer are used as the input gas sensor signals for the learning models. To align with the spatial resolution used in our simulated training data, the raw measurements are interpolated onto a 64×64 grid, forming dense ground truth maps of gas concentration and wind fields. From these maps, sparse sensor inputs are generated using the same probabilistic sampling strategies



(a) Experimental environment with active gas source.



(b) Vertical arrangement of 4 sensing units.



(c) Time series of MOX and PID responses at Layer 1.

Fig. 4: The experimental setup of the low-speed wind tunnel dataset used in our study [33]. (A): An example of the experimental environment with an active gas source; (B): Vertical arrangement of 4 sensing units at different heights. (C): Time series of MOX and PID sensor responses at Layer 1 in the real-world dataset.

defined in Eq. 5, ensuring consistency in the sampling process between training and evaluation. The ad hoc evaluation setup based on the real-world dataset allows a realistic and challenging benchmark to assess the considered approaches under practical conditions, featuring nonuniform sampling, real sensor noise, and complex airflow dynamics.

D. Baseline Methods and Evaluation Criteria

To assess the effectiveness of the proposed method, we conducted quantitative comparisons against several

baseline approaches: Linear Interpolation, Kernel DM+V, and a CNN-based encoder-decoder model.

- Linear Interpolation serves as a straightforward baseline. It estimates gas concentrations across the spatial domain by interpolating between known sensor observations using spatial proximity.
- Kernel Extrapolation Distribution Mapping + Variance (Kernel DM+V) [16] is a non-parametric method that models the gas distribution as a probabilistic field. It uses Gaussian kernels to extrapolate measurements and compute variance estimates, capturing both concentration and uncertainty from sparse sensor data.
- CNN encoder-decoder model is a learning-based baseline adapted from the state-of-the-art work [21]. It transforms sparse sensor inputs into dense prediction maps via hierarchical feature extraction and upsampling.

To ensure fair comparison, we constructed a CNN model with a similar number of parameters to our transformer model. It takes a 4-channel input of spatial resolution 64×64 , and outputs a 3-channel prediction map of the same size. Since CNNs are not able to handle variable-length inputs, the sensor observations were first zero-padded based on their spatial coordinates to form a fixed-size 4-channel matrix of size 64×64 . The four channels represent gas concentration, wind velocities (u, v), and a binary mask to indicate sensor locations. The CNN architecture uses three convolutional layers for encoding and two transposed convolutional layers, followed by a final convolution for decoding. The encoder progressively reduces spatial resolution ($64 \rightarrow 32 \rightarrow 16$) while increasing feature depth to capture high-level representations. LeakyReLU activations are used throughout the network to prevent inactive neurons.

We evaluated the models on a held-out test set using two quantitative metrics, namely Mean Squared Error (MSE) and Structural Similarity Index (SSIM).

- MSE quantifies the average squared difference between predicted values and ground truth, which is given by

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (6)$$

- SSIM captures the perceptual similarity between predicted and ground truth spatial fields, considering local structural patterns [34]. SSIM is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (7)$$

where μ_x, μ_y are local means, σ_x^2, σ_y^2 are variances, and σ_{xy} is the covariance. C_1 and C_2 are small constants to stabilize division. SSIM ranges from 0 (no similarity) to 1 (perfect match).

TABLE II: Evaluation of Different Approaches

Sampling Condition	Reconstructed Map	Linear Interpolation		Kernel DM+V		CNN		Transformer	
		MSE	SSIM	MSE	SSIM	MSE	SSIM	MSE	SSIM
100 Samples	Gas	0.0400	0.5308	0.0090	0.6809	0.0095	0.5560	0.0003	0.9830
	Wind	0.0098	0.6186	0.0023	0.8601	0.0011	0.8276	0.0001	0.9733
50 Samples	Gas	0.0386	0.4932	0.0135	0.6664	0.0125	0.4851	0.0003	0.9822
	Wind	0.0115	0.5710	0.0063	0.7328	0.0019	0.7928	0.0001	0.9724
20 Samples	Gas	0.0382	0.4669	0.0243	0.6473	0.0202	0.2875	0.0005	0.9767
	Wind	0.0165	0.4609	0.0186	0.4820	0.0076	0.5945	0.0002	0.9676
50 Plume-favoured Samples ($\alpha = 0.5$)	Gas	0.0118	0.6090	0.0464	0.4046	0.0358	0.2119	0.0004	0.9749
	Wind	0.0094	0.6418	0.0137	0.5132	0.0052	0.5989	0.0003	0.9677
50 Plume-avoiding Samples ($\alpha = -0.5$)	Gas	0.0327	0.6464	0.0403	0.5844	0.0522	0.2401	0.0008	0.9675
	Wind	0.0097	0.7045	0.0151	0.5125	0.0064	0.6514	0.0002	0.9654

Importantly, the simulation configurations in the test set are entirely distinct from those used during training, ensuring that no shared gas source or wind field scenario appears in both sets.

To thoroughly evaluate model performance under varying observation conditions, we define two testing scenarios: Sparsity-level variation and sampling bias.

- Sparsity-level variation: The number of input observations is varied (100, 50, 20 samples per case). For each sparsity level, MSE and SSIM are averaged across the test set to assess how model performance degrades with reduced sampling.
- Sampling bias: The number of samples is fixed at 50, but spatial sampling bias is introduced by following the sampling strategy defined by Eq. 5. We evaluate model performance for $\alpha = 0.5$ (relatively plume-favoured) and $\alpha = -0.5$ (relatively plume-avoiding) using MSE and SSIM. Note that $\alpha = 0$ overlaps with the 50-sample case from the sparsity evaluation.

IV. Results

We quantitatively evaluate all models using the held-out test set, with performance summarized in Table II. Overall, across all sensor sparsity levels and sampling biases, our transformer-based model consistently achieves the lowest mean squared error (MSE) and highest structural similarity (SSIM) for both gas concentration and wind field reconstruction. Compared to baseline methods, including linear interpolation, Kernel DM+V, and CNN, the transformer demonstrates significantly higher robustness to reduced sample counts and biased sampling. The subsequent Section IV-A provides a more detailed breakdown of these results under varying sparsity levels and sampling biases, along with qualitative visual comparisons based on representative cases. In Section IV-B, we show that our model produces a qualitatively realistic gas plume reconstruction with real wind-tunnel data. This result reflects our work’s potential in sim-to-real generalization.

A. Evaluation with Simulated Ground-truth

1) Performance under the varied sampling sparsity levels: From Table II, one can note that even under the most extreme sparse sensor conditions (i.e., using 20 samples), our work still achieves significantly lower MSE and higher SSIM scores for both gas and wind field reconstructions, indicating its ability to minimize error while preserving spatial structures of the wind field and gas distribution. In contrast, methods like Kernel DM+V and linear interpolation exhibit their key limitation, i.e., the reliance on dense local data, which is reflected by a substantial performance degradation as observations decrease from 100 to 50, and then to 20 samples. CNNs provide moderate adaptability but still struggle with capturing long-range dependencies. Leveraging global attention mechanisms, our transformer-based model can effectively integrate spatial information from limited and irregular inputs, maintaining both local detail and overall coherence even in low-observation settings.

To complement the quantitative results, Figure 5 provides a comparison of the reconstructed gas distribution maps and wind fields under different sampling sparsity levels. As shown, when the number of sensor observations is insufficient to evenly cover the entire plume region, linear interpolation tends to overestimate the values between positive samples. This results in unrealistic plume boundaries and expansion of the gas region. In contrast, the Kernel DM+V method underestimates concentrations in unsampled plume areas, often leading to hollow or fragmented reconstructions. The CNN encoder-decoder, while more robust, exhibits increasingly blurred predictions as the sample count decreases. This blurring effect likely reflects a bias toward the average patterns observed in the training data, causing the model to regress toward conservative outputs in the absence of sufficient local information.

2) Performance with sampling bias: From Table II, under the biased sampling condition, i.e., given 50 samples acquired by a plume-avoiding sampling strategy, the transformer continues to outperform all other methods.

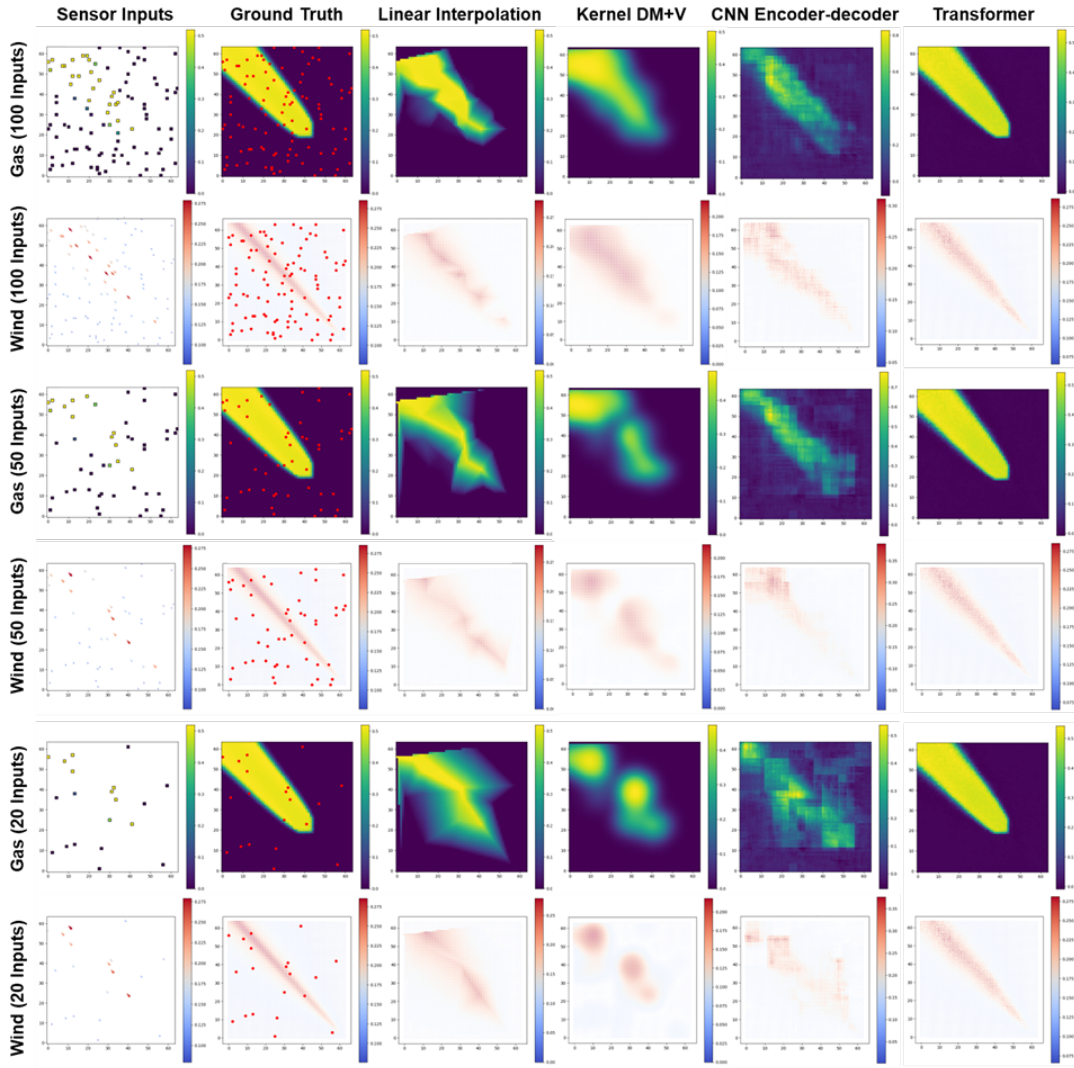


Fig. 5: Comparison of different approaches under different sampling sparsity levels. The figure presents reconstructed gas concentration and wind fields for the same test sample using Linear Interpolation, Kernel DM+V, CNN, and the proposed Transformer-based model. As sensor observations become sparser, baseline methods exhibit significant degradation, while our work preserves both spatial accuracy and structural coherence.

While linear interpolation performs reasonably well in densely sampled regions, it fails to reconstruct transitions in sparsely covered areas. Kernel DM+V and CNN models exhibit reduced generalization when training data are unevenly distributed, resulting in higher reconstruction errors and lower SSIM values. Although the transformer’s performance also slightly declines when the plume region is under-sampled, it maintains much greater structural fidelity and lower error rates. Its ability to globally reason across sampled and unsampled regions enables it to deliver stable and accurate reconstructions, even when the input distribution is highly biased.

Figure 6 illustrates the comparison of the reconstructed gas distribution maps and wind fields when sampling

biases are posed. When sensor observations are denser within the plume region (acquired by a plume-favoured sampling strategy), both linear interpolation and Kernel DM+V tend to overestimate the extent of the plume, resulting in some exaggeration of the gas region. Nonetheless, in these scenarios, they are still able to provide reasonable coverage of the overall concentration field. However, when the plume region is sparsely sampled (acquired by a plume-avoiding sampling strategy), both methods show significant distortion. The dominance of negative samples causes them to under-represent or miss the plume. The CNN model, on the other hand, demonstrates a tendency to regress toward the mean prediction under both dense and sparse sampling conditions. This produces

blurred outputs and fails to capture the sharp spatial structure of the plume. This behaviour suggests that CNN may struggle to model long-range spatial dependencies, making it less capable of adapting to unstructured sensor inputs.

B. Evaluation with Real-world Data

To assess the model’s generalization ability and move towards addressing the sim-to-real gap, we further evaluate our work and the base methods with the real-world wind tunnel data (presented in Section III-C). Due to the absence of high-resolution ground truth, we conducted a qualitative case study analysis. To validate the fidelity of our work, the model outputs were generated directly from raw, uncalibrated photo ionization sensor readings without applying any post-processing or enhancement. The only adjustment was a linear scaling of the sensor values to align them with the dynamic range of the simulation-based training data.

A comparison of the reconstructed gas distribution map is given by the first row of Figure 7A. Compared to the actual experimental setup shown in Figure 7B, our work managed to capture most of the gas plume in the reconstruction, outperforming other methods, which did not reflect the presence of the gas plume.

In our view, the ad hoc real-world data is challenging for gas distribution mapping due to the low signal-to-noise ratios in gas sampling. During the sequential sampling process from upwind to downwind, the sensor exhibited noticeable baseline drift, resulting in a systematically elevated background concentration in the upper half of the observed distribution. As a consequence, the actual gas plume, highlighted by the red box in Fig. 7B, which corresponds to the visible plume region in the experimental area, was submerged within the background signal. Under these challenging real-world conditions, methods that are sensitive to the gas sensor baseline, such as linear interpolation, Kernel DM+V, and CNN misinterpreted the influence range of the plume due to the background offset, leading to inaccurate reconstructions. In contrast, the proposed transformed-based approach successfully reconstructed a complete plume region despite the strong background drift. Its global attention mechanism allowed it to distinguish the true signal pattern from the noise-dominated observation, effectively identifying the plume shape and orientation. However, the transformer’s output was still affected by the noise during measurement; the reconstructed plume contours did not fully align with the ground truth, and two overlapping plume heads appeared in the upwind region. These reconstruction errors likely stem from distorted wind direction observations, indicating that despite the model’s robustness, a sim-to-real gap remains.

Regarding the wind field reconstruction, as shown in the second row of Figure 7A, none of the methods were able to accurately reconstruct the wind field distribution. This

limitation is primarily attributed to the presence of obstacles within the test area, which introduced numerous low-speed or turbulent regions. These complex flow dynamics disrupt the spatial continuity of the wind field, making it difficult for numerical methods to interpolate or infer missing information from sparse observations. Moreover, in the absence of prior knowledge about the location or shape of obstacles, and without having encountered similar flow patterns during training, the models lacked the environment-informed contextual information necessary to predict the local disturbances caused by the obstacles.

V. Discussion and Conclusion

Gas distribution mapping (GDM) in mobile robotics has long faced the challenge of generating accurate spatial reconstructions from sparse and uneven sensor data. Traditional approaches often rely on dense sampling or assume idealized plume models, which rarely hold in complex real-world environments. Learning-based methods, especially convolutional networks, have improved reconstruction performance, but they typically struggle to handle highly irregular inputs or to capture long-range spatial dependencies. These gaps underscore the need for models that are both structurally flexible and data-efficient.

In this work, we present a transformer-based model for GDM that directly addresses these limitations. By leveraging attention mechanisms and positional encoding, the model learns to globally integrate sparse sensor measurements and infer spatial structure from limited and biased samples. Quantitative evaluations on synthetic test cases demonstrate the robustness of the approach: under extremely sparse sampling (20 observations), the transformer achieves an MSE of 0.0005 and SSIM above 0.96 for gas concentration prediction, substantially outperforming the baseline methods, linear interpolation, Kernel DM+V, and CNN-based models. Even under the sampling bias, where sensor inputs are intentionally skewed away from the plume, our work maintains strong reconstruction fidelity, while the compared baseline methods deteriorate significantly. In the real-world wind tunnel experiment, the transformer model demonstrated an advantage in handling background noise caused by sensor baseline drifting. While other methods misinterpreted the elevated background as part of the plume, the transformer-based approach effectively separated the true plume region from the noisy observations.

These findings reinforce the emerging consensus in spatial modelling that transformers offer appealing advantages over convolutional deep learning models, particularly for unstructured inputs and tasks involving long-range correlation. Whereas CNNs tend to regress toward averaged patterns seen during training, the transformer is able to preserve both local plume features and broader spatial continuity, leading to more faithful reconstructions across a variety of input conditions. Compared to the data-driven

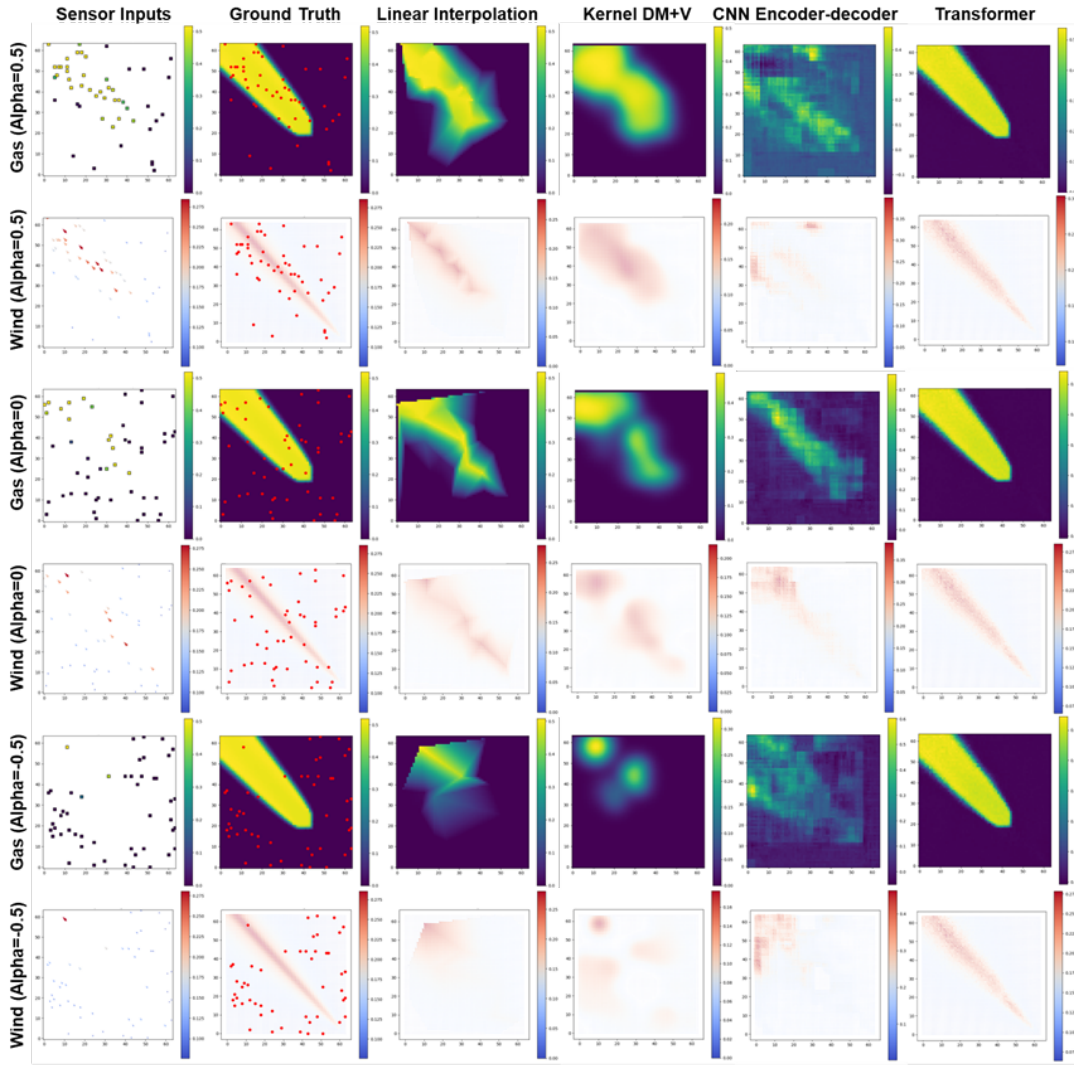


Fig. 6: comparison of different approaches under biased sampling conditions. The figure presents reconstructed gas concentration and wind fields for the same test sample using Linear Interpolation, Kernel DM+V, CNN, and the proposed Transformer-based model. When sensor inputs are biased toward or away from the gas plume region, baseline methods either exaggerate or miss critical plume structures, whereas the Transformer model maintains accurate and coherent reconstructions.

Kernel DM+V, which assumes smooth spatial extrapolation and is sensitive to sampling bias, the transformer is markedly more adaptive.

However, several limitations should be acknowledged. First, our work still exhibits a sim-to-real gap. The transformer-based model is trained entirely on simulated data, and although it is evaluated on real wind tunnel measurements, the simulation-to-reality transfer remains imperfect. In real-world environments, obstacles introduce complex wind field distortions that are not represented in the training data. Simple noise injections during training are insufficient to capture such structural irregularities. Furthermore, due to the presence of cross-attention between wind and concentration channels, inaccuracies in

wind prediction propagate to gas concentration estimates, causing distortions in the reconstructed plume shape and location.

Apart from the observed limitation in the wind field reconstruction, a potential concern is that the model might implicitly memorise typical plume patterns from the training data, and its success on the test set could partly stem from structural similarities between training and evaluation scenarios. Furthermore, the transformer operates in a data-driven manner without explicitly enforcing physical laws such as mass conservation or fluid dynamics, which could limit its generalization under turbulent or multi-source gas dispersion.

The observed and potential limitations suggest that

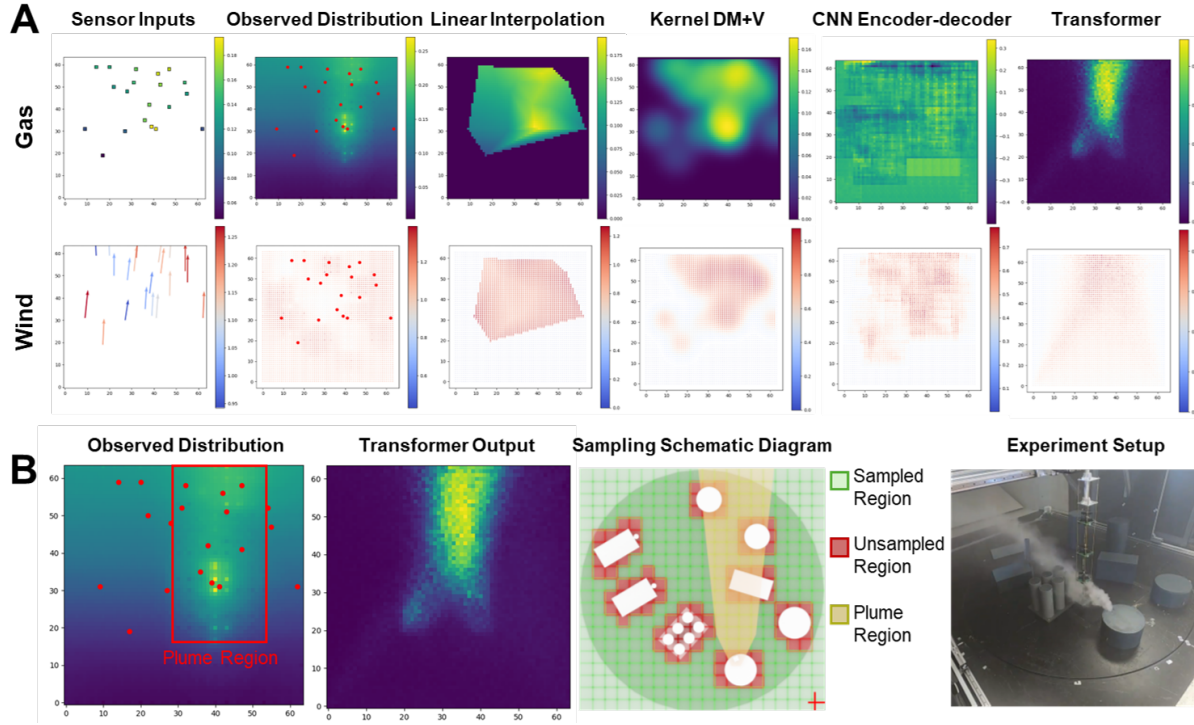


Fig. 7: Evaluation with real-world data. (A) Models' performance on data from wind tunnel experiments; (B) Comparison of the dense observed gas distribution map, the reconstruction by our work, and the experimental setup with the real plume region.

while the transformer offers a powerful framework for data-driven GDM, future efforts are needed to address the generalisation to real-world scenarios [35]. Future work should improve the training data in terms of expanding the diversity of training scenarios, including more complex environmental geometries and variable gas dispersion patterns [36]. In addition, integrating physics-informed training approach [37] and domain-specific inductive biases could allow the model to be better aware of the underlying fluid dynamics. However, physics-informed training has higher requirement to data quality, as noise in training may cause gradient between physics loss and dataset loss, resulting in trivial solution. To improve robustness, especially to incorporate with widely used metal-oxide gas sensors, noise and sensor response models that more accurately reflect the behaviour of real gas sensors should be considered. Constructing such training datasets will likely require a hybrid approach that combines high-fidelity simulations with real-world experimental data.

Additionally, deploying the model on a mobile robotic platform in outdoor or industrial environments would enable real-time evaluation under realistic conditions and provide insights into practical deployment, resilience to sensor drift, and adaptive planning for efficient sampling. Besides, extending the framework to three-dimensional environments is a natural but important future direction towards realistic, complex applications. In many practical

scenarios, such as multi-floor buildings or underground tunnels, gas dispersion occurs in all three spatial dimensions. A 3D extension of GDM would enable more comprehensive monitoring and improved applicability to real-world deployments.

Acknowledgment

This work has been partially supported by the German Federal Ministry of Research, Technology and Space (BMFTR) under the Robotics Institute Germany (RIG).

CRediT authorship contribution statement

Changhao Tian: Writing, Investigation, Hongyuan Di: Investigation, Methodology, Han Fan: Writing, Reviewing, Chaojun Wang: Supervision, Achim J. Lilienthal: Supervision, Funding acquisition, Reviewing

Competing interests

The authors declare no competing interests.

Data availability

The dataset used in this study is openly accessible at our GitHub repository: GitHub repository.

Code availability

All training scripts, model evaluation Jupyter notebooks used to generate the figures, and the trained model checkpoint are openly accessible at our GitHub repository: GitHub repository. All code and the trained model are released under the MIT license.

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